3 Sensing the real world

This chapter covers

- The role of sensors in updating a digital twin
- Building a complete sensing system for real-world applications
- Processing sensor data
- Managing sensors at scale

Digital twins depend on maintaining an accurate and timely connection to reality. While 3D models show you what a system looked like when designed, a true digital twin evolves continuously with its physical counterpart through systematic sensing of the physical world. Sensors bridge the physical and digital worlds, continuously capturing changing conditions such as temperature fluctuations, equipment vibrations, fluid flows, and human activities and converting them into data streams that keep digital twins synchronized with reality.

This data foundation allows digital twins to not just monitor, but to optimize system performance by adjusting parameters in the real system based on simulated outcomes. As digital twins mature from descriptive (what happened) through predictive (what will happen) to autonomous capabilities (automatically adjusting system behavior), the demands for rich, timely data increase dramatically.

This chapter walks you through building a complete sensing architecture using the home digital twin project to illustrate common challenges in any digital twin project: selecting appropriate sensors, handling infrastructure constraints, and balancing cost with capability. You'll learn how to design sensing systems that capture the right data at the right frequency, connect diverse sensors through multiple communication protocols, and process sensor data into formats suitable for digital twin applications. By the end of this chapter, you'll understand how to architect sensing systems that provide the data foundation necessary for digital twin applications across any domain.